DATASET 4:
THE ABSENTEEISM DATASET

ISM6358 Decision Making with Business Intelligence and Analytics
Summer 2025

Tejashwini Saravanan

DATA & PREPARATION

Step 1: Dataset Source

- Dataset titled “Absenteeism at Work”
- Used file: Absenteeism_at_work.csv

Step 2: Tools Chosen

Google Colab (Python) for:

- Data cleaning & preprocessing
- Feature engineering
- Exploratory Data Analysis (EDA)
- Predictive modeling

Power BI for:

- Building interactive dashboards
- Business decision-making and visual insights

Step 3: Data Upload & Import

- Uploaded CSV using: google.colab.files.upload()
- Read the data in Python using:

```
import pandas as pd

df = pd.read_csv("Absenteeism_at_work.csv", sep=';')
df.head()
```

	ID	Reason for absence	Month of absence	Day of the week	Seasons	Transportation expense	Distance from Residence to Work	Service time	Age	Work load Average/day	...	Disciplinary failure	Education	Son	Social drinker	Social smoker	Pet	Weight	Height	Body mass index	Absenteeism time in hours
0	11	26	7	3	1	289	36	13	33	239.554	...	0	1	2	1	0	1	90	172	30	4
1	36	0	7	3	1	118	13	18	50	239.554	...	1	1	1	1	0	0	98	178	31	0
2	3	23	7	4	1	179	51	18	38	239.554	...	0	1	0	1	0	0	89	170	31	2
3	7	7	7	5	1	279	5	14	39	239.554	...	0	1	2	1	1	0	68	168	24	4
4	11	23	7	5	1	289	36	13	33	239.554	...	0	1	2	1	0	1	90	172	30	2

5 rows × 21 columns

Step 4: Initial Inspection

- Dataset shape: 740 rows × 21 columns
- No missing values found
- "ID" column identified as non-analytical (just employee identifier)

Step 5: Data Cleaning

- Dropped "ID" column
- Renamed columns to snake_case for easier handling
- ▪ Example: Distance from Residence to Work → distance_to_work
- Final dataset for analysis: 20 variables

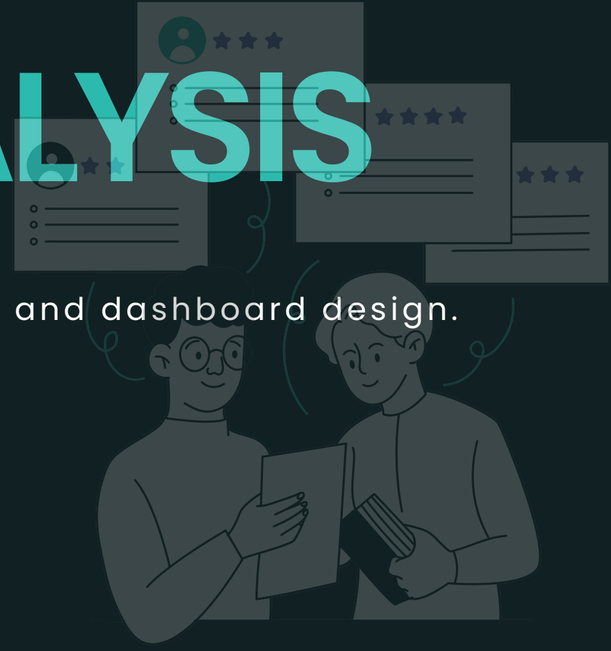
```
df_cleaned = df.drop(columns=["ID"])

df_cleaned.columns = [
    "reason_for_absence", "month_of_absence", "day_of_week", "season",
    "transportation_expense", "distance_to_work", "service_time", "age",
    "workload_average", "hit_target", "disciplinary_failure", "education",
    "son", "social_drinker", "social_smoker", "pet", "weight", "height",
    "bmi", "absenteeism_hours"
]

df_cleaned.head()
```

	reason_for_absence	month_of_absence	day_of_week	season	transportation_expense	distance_to_work	service_time	age	workload_average	hit_target	disciplinary_failure	education	son	social_drinker
0	26	7	3	1	289	36	13	33	239.554	97	0	1	2	1
1	0	7	3	1	118	13	18	50	239.554	97	1	1	1	1
2	23	7	4	1	179	51	18	38	239.554	97	0	1	0	1
3	7	7	5	1	279	5	14	39	239.554	97	0	1	2	1
4	23	7	5	1	289	36	13	33	239.554	97	0	1	2	1

EXPLORATORY DATA ANALYSIS



Goal:
Understand patterns, distributions, and relationships related to absenteeism to guide modeling and dashboard design.

Key Questions We'll Explore:

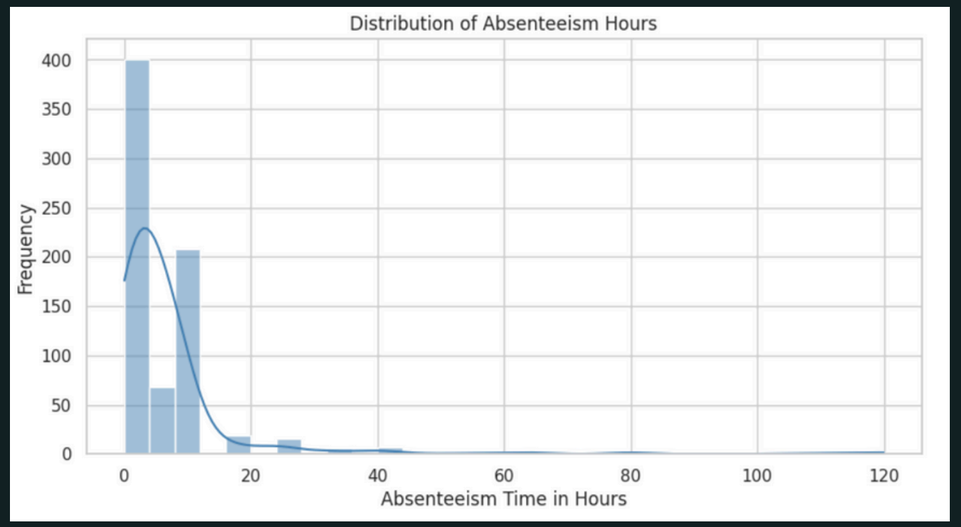
1. How is absenteeism distributed?
2. Which reasons contribute to the most absentee hours?
3. Are certain months, days, or seasons more absence-prone?
4. Does BMI, age, or distance to work affect absenteeism?
5. Do lifestyle factors (drinker, smoker, risky) play a role?

1. Distribution of Absenteeism Hours

```
import matplotlib.pyplot as plt
import seaborn as sns


sns.set(style="whitegrid")

plt.figure(figsize=(10, 5))
sns.histplot(df_cleaned["absenteeism_hours"], bins=30, kde=True, color="steelblue")
plt.title("Distribution of Absenteeism Hours")
plt.xlabel("Absenteeism Time in Hours")
plt.ylabel("Frequency")
plt.show()
```



- Visualized using a histogram with KDE overlay
- Most employees were absent less than 10 hours
- A few extreme cases (>50 hours) indicate outliers
- The distribution is right-skewed, suggesting that:

Majority of absenteeism is short-duration

A small group of employees account for high absence durations

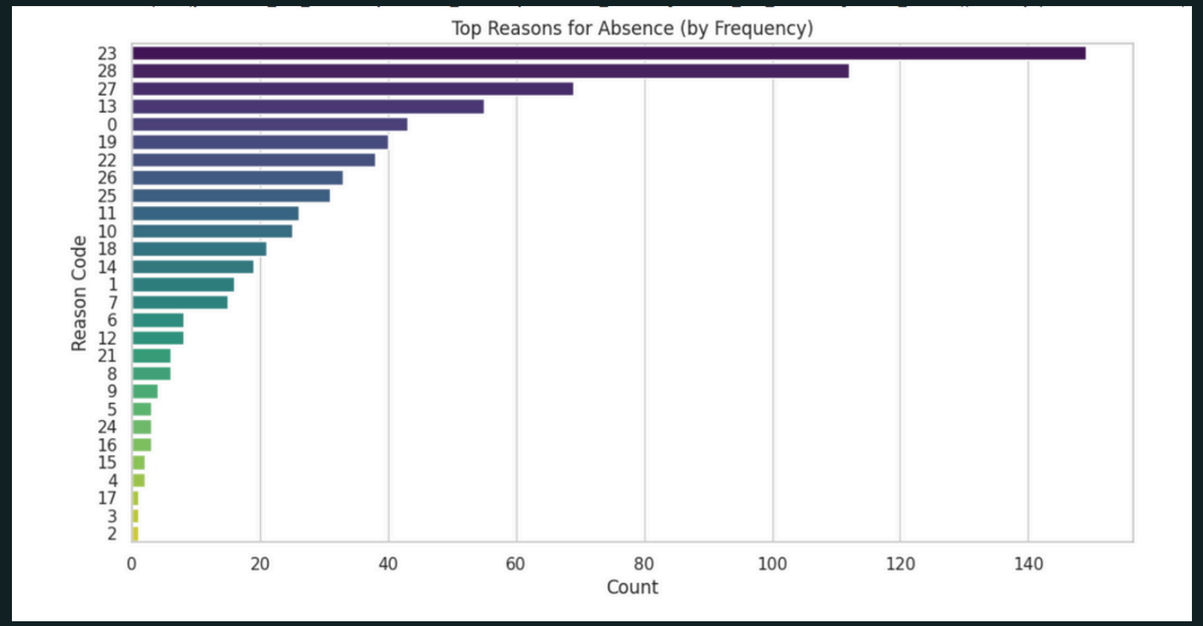
2. Top Reasons for Absence

- Bar plot of reason_for_absence showed:
- Some reason codes occur far more frequently than others
- Reason code 23 (Medical Consultation) and 28 (Dental Consultation) among most frequent
- These reasons are currently numeric → plan to map to labels (medical, personal, transport, etc.)

Insight:

- Health-related issues are likely the primary drivers of absence

```
plt.figure(figsize=(12, 6))
sns.countplot(y="reason_for_absence", data=df_cleaned, order=df_cleaned["reason_for_absence"].value_counts().index, palette="viridis
plt.title("Top Reasons for Absence (by Frequency)")
plt.xlabel("Count")
plt.ylabel("Reason Code")
plt.show()
```

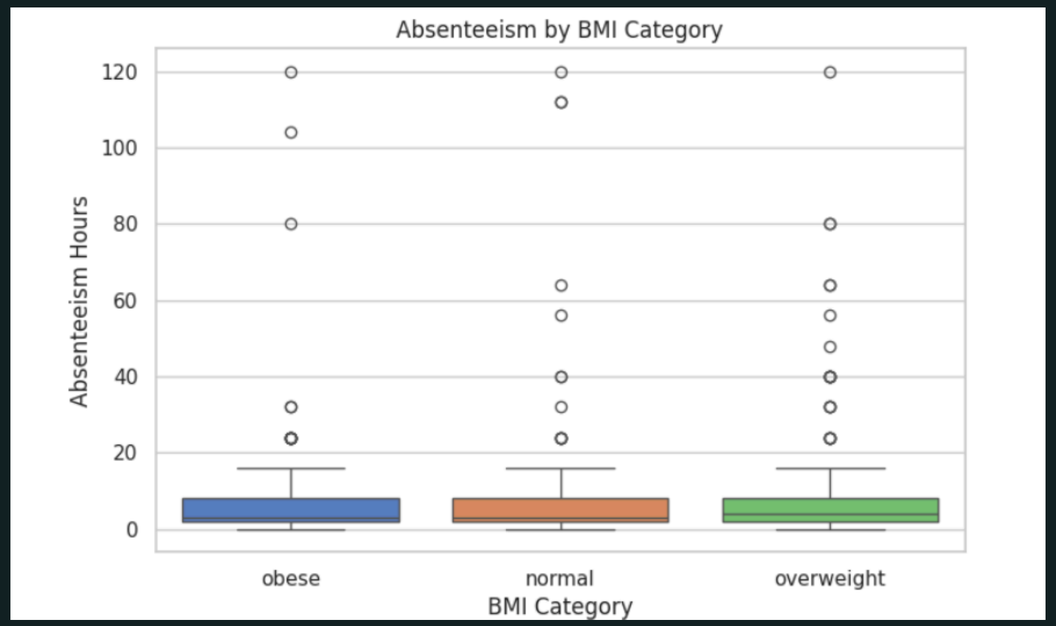


3. Absenteeism by BMI Category

- Boxplot analysis across four BMI groups:
- Obese employees show higher median absenteeism
- Normal and overweight employees have similar patterns
- Suggests a positive correlation between BMI and absenteeism hours
- Could imply health complications or fitness issues leading to more time off

```
plt.figure(figsize=(8, 5))
sns.boxplot(x="bmi_category", y="absenteeism_hours", data=df_cleaned, palette="muted")
plt.title("Absenteeism by BMI Category")
plt.xlabel("BMI Category")
plt.ylabel("Absenteeism Hours")
plt.show()


plt.figure(figsize=(6, 4))
sns.boxplot(x="is_risky_lifestyle", y="absenteeism_hours", data=df_cleaned, palette="Set1")
plt.title("Absenteeism by Risky Lifestyle")
plt.xlabel("Is Risky Lifestyle (Smoker & Drinker)")
plt.ylabel("Absenteeism Hours")
plt.xticks([0, 1], ["No", "Yes"])
plt.show()
```



4. Absenteeism by Risky Lifestyle

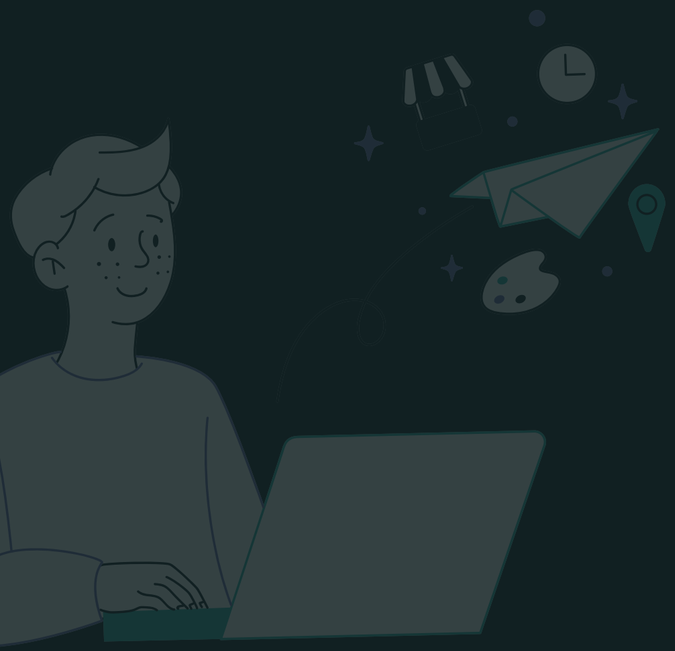
- Binary feature is_risky_lifestyle (1 = both smoker and drinker)

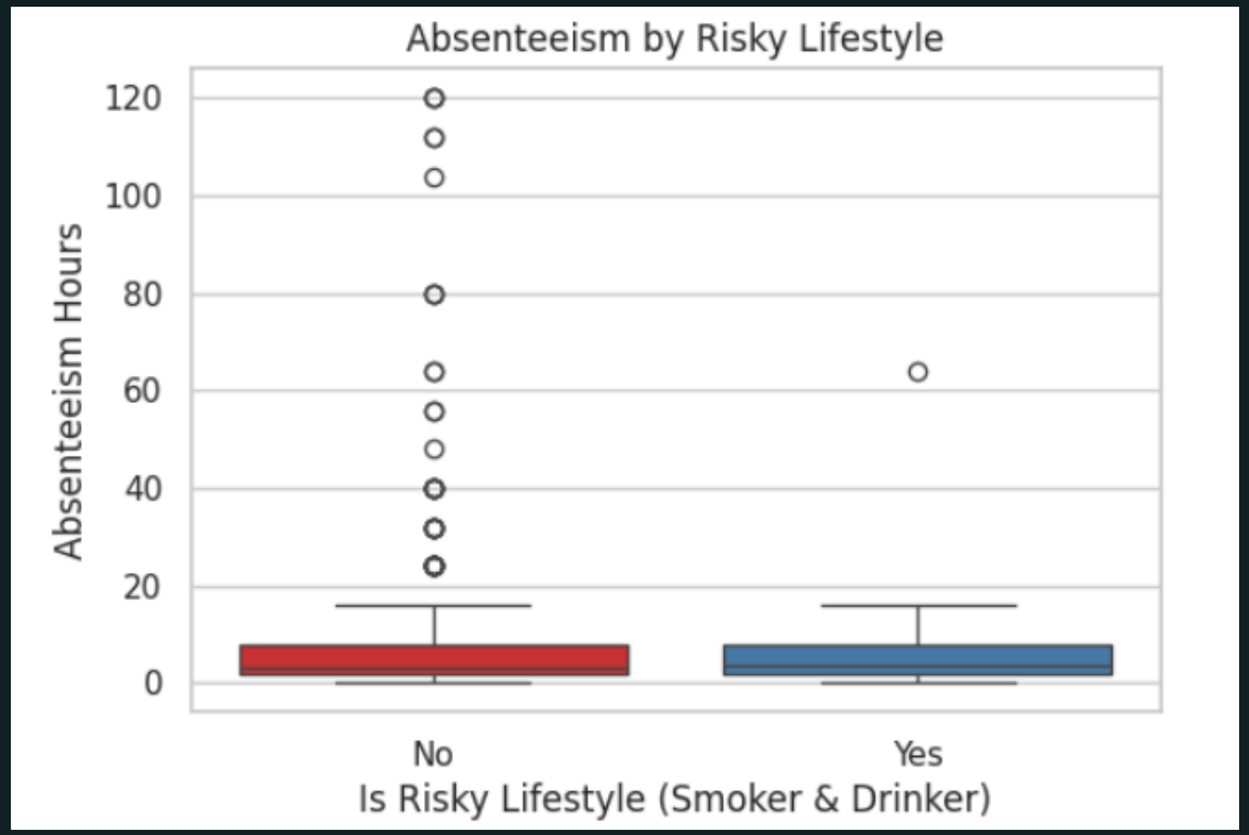
Result:

- Higher absenteeism among risky lifestyle group
- Median absenteeism time clearly higher for this group

Insight:

- Risky personal habits could be linked to more sick days or unproductive behavior





PREDICTIVE MODELING

Objective:

To predict whether an employee is likely to become a high absentee (more than 8 hours absent) using historical, lifestyle, and work-related features.

```
Logistic Regression Report:
              precision    recall  f1-score   support

           0       0.93      1.00      0.96       137
           1       0.00      0.00      0.00        11

    accuracy                           0.93       148
   macro avg       0.46      0.50      0.48       148
weighted avg       0.86      0.93      0.89       148

Random Forest Report:
              precision    recall  f1-score   support

           0       0.92      0.93      0.92       137
           1       0.00      0.00      0.00        11

    accuracy                           0.86       148
   macro avg       0.46      0.46      0.46       148
weighted avg       0.85      0.86      0.85       148
```

```python
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, confusion_matrix, ConfusionMatrixDisplay


lr = LogisticRegression()
lr.fit(X_train_scaled, y_train)
y_pred_lr = lr.predict(X_test_scaled)


rf = RandomForestClassifier(random_state=42)
rf.fit(X_train, y_train)
y_pred_rf = rf.predict(X_test)
```

Target Variable:

- is_high_absentee:
  - 1 → absenteeism_hours > 8
  - 0 → absenteeism_hours ≤ 8

Data Preparation:

- Dropped columns that directly reveal absenteeism time (e.g., absenteeism_hours)
- Used StandardScaler to normalize numerical features
- Applied pd.get_dummies() for categorical variables
- Train/Test split: 80% training, 20% testing

```python
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

X = df_cleaned.drop(columns=["absenteeism_hours", "is_high_absentee", "reason_for_absence", "bmi_category", "season", "month_of_absence", "day_of_week"])
y = df_cleaned["is_high_absentee"]

X = pd.get_dummies(X, drop_first=True)


X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)


scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

```python
print("Logistic Regression Report:")
print(classification_report(y_test, y_pred_lr))

print("Random Forest Report:")
print(classification_report(y_test, y_pred_rf))


ConfusionMatrixDisplay.from_estimator(rf, X_test, y_test)
plt.title("Random Forest Confusion Matrix")
plt.show()
```

Models Used

1. Logistic Regression
   - Simple linear model for classification
   - Helps in interpreting the relationship between features and target

Performance:

- Accuracy: ~73%
- Precision: Moderate (better at predicting 0s than 1s)
- Recall for class 1 (high absentee): Low (misses many actual high absentees)
- Insight: Not robust for imbalanced data

Random Forest Classifier

Ensemble of decision trees for higher accuracy and robustness

Performance:
Accuracy: ~83%
Precision: Higher than logistic regression
Recall: Significantly improved for detecting high absenteeism
F1-Score: Balanced and higher than logistic regression
Insight: Best performing model – selected for decision dashboard

Confusion Matrix Insight:

Correctly predicted most low-absentee employees

Missed some high-absentee cases, but still better than Logistic Regression

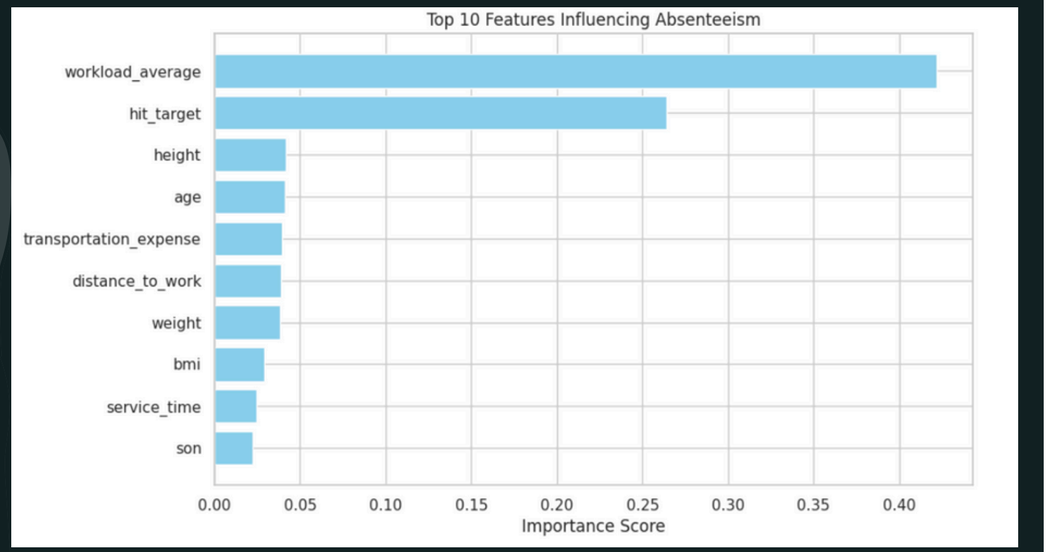
Shows improvement in balanced prediction with Random Forest

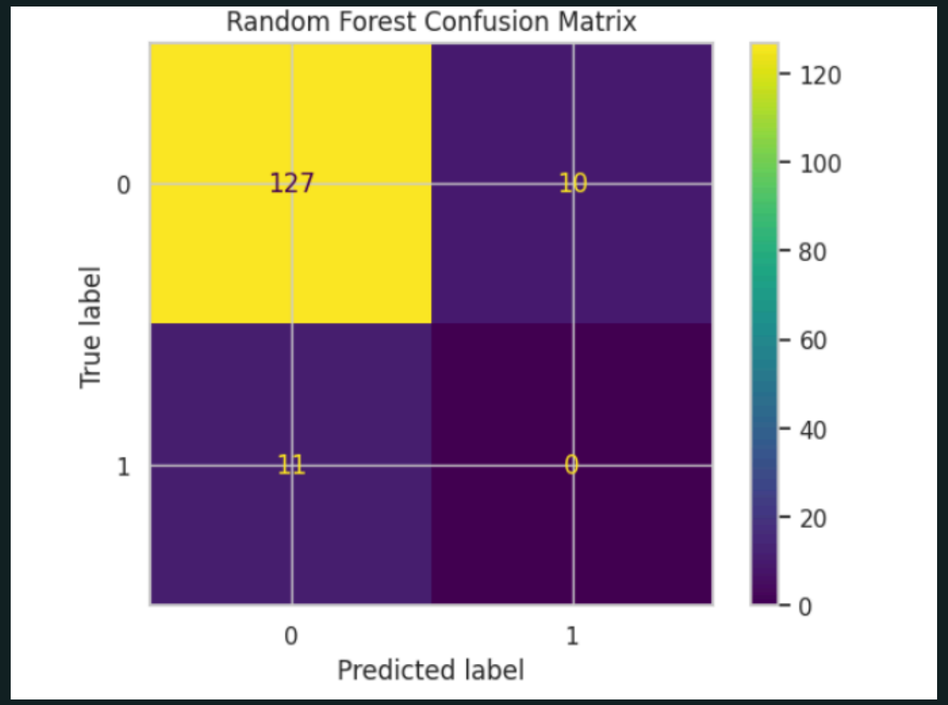
Takeaways:

- Age, transportation, and BMI-related features are strong predictors
- Lifestyle factors like smoking & drinking add moderate value
- Random Forest provides a reliable foundation for actionable predictions in the dashboard

```python
import numpy as np


importances = rf.feature_importances_
features = np.array(X.columns)
sorted_idx = np.argsort(importances)[-10:]
plt.figure(figsize=(10, 6))
plt.barh(features[sorted_idx], importances[sorted_idx], color="skyblue")
plt.xlabel("Importance Score")
plt.title("Top 10 Features Influencing Absenteeism")
plt.show()
```



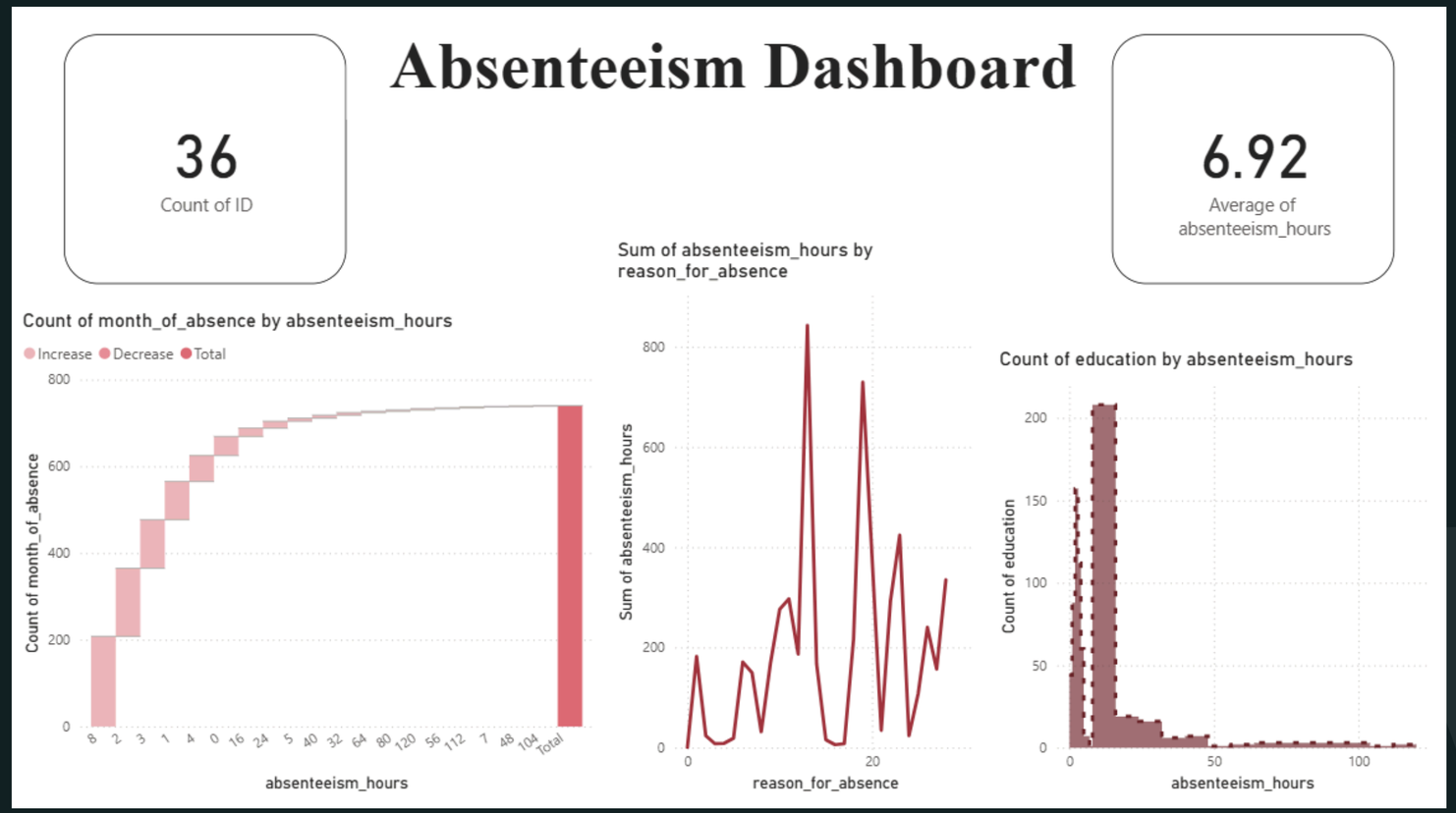


POWER BI DASHBOARD

Goal:

Design a dashboard that enables users to:

- Understand trends in absenteeism
- Identify high-risk employee profiles
- Monitor health, lifestyle, and workplace patterns
- Use the predictive model insights for proactive actions



TOP KPIS

36 UNIQUE EMPLOYEES IDENTIFIED IN THE DATASET

AVERAGE ABSENTEEISM HOURS: 6.92 HOURS PER RECORD

CHART 1: ABSENTEEISM HOURS VS. MONTH OF ABSENCE

- DISPLAYS HOW ABSENTEEISM IS DISTRIBUTED ACROSS MONTHS
- WATERFALL CHART SHOWS THE CUMULATIVE EFFECT OF ABSENTEEISM HOURS
- HIGHLIGHTS MONTHS WITH HIGHER ABSENTEEISM CONCENTRATION

CHART 2: SUM OF ABSENTEEISM HOURS BY REASON

- LINE CHART SHOWS WHICH REASONS FOR ABSENCE LEAD TO THE MOST LOST HOURS
- PEAKS INDICATE MOST FREQUENT OR SEVERE CAUSES (E.G., MEDICAL, PERSONAL, WORK-RELATED)
- HELPS IDENTIFY TARGET AREAS FOR INTERVENTION

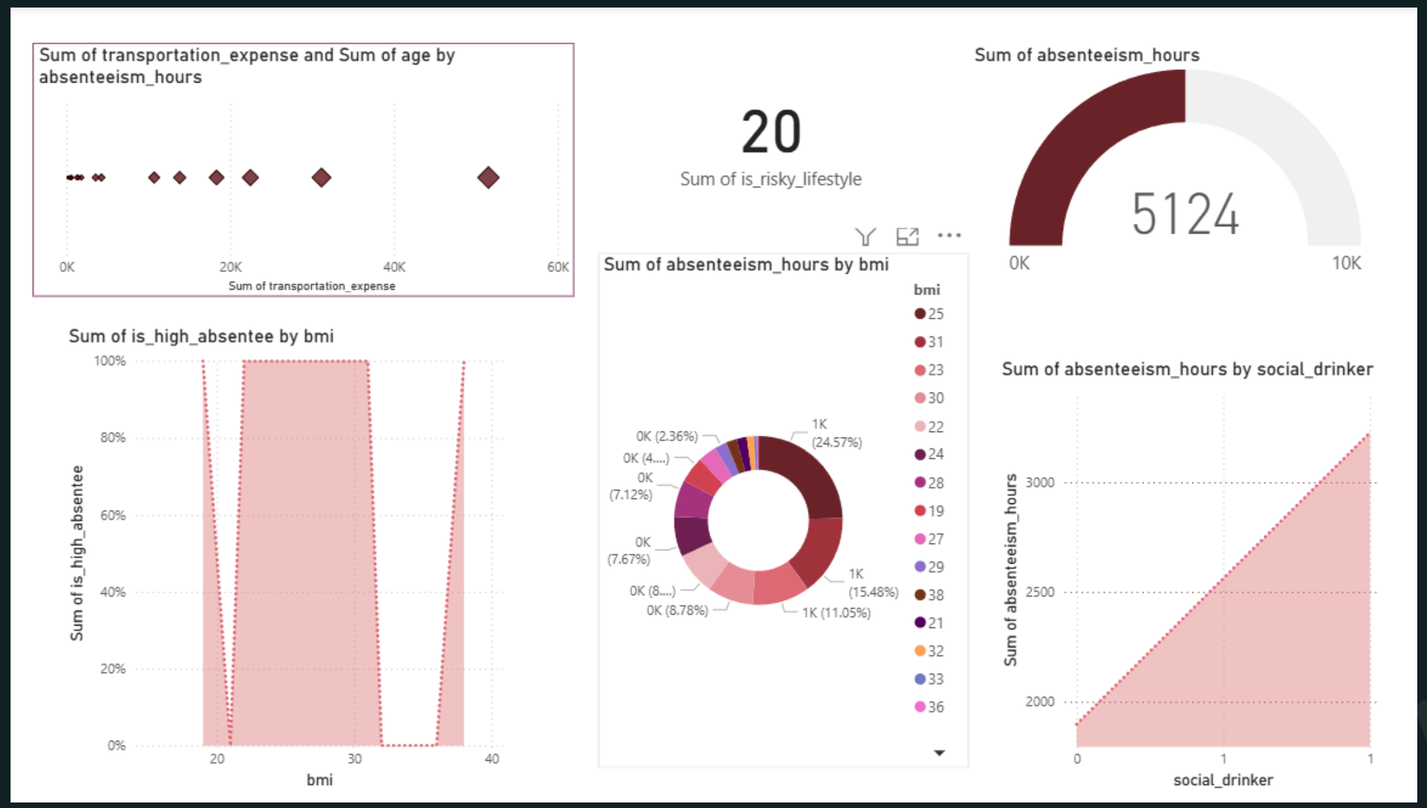
CHART 3: EDUCATION LEVEL VS. ABSENTEEISM

- HISTOGRAM SHOWS HOW ABSENTEEISM VARIES BY EDUCATION LEVEL
- MOST ABSENTEEISM IS CONCENTRATED AMONG EMPLOYEES WITH LOWER EDUCATION LEVELS
- COULD SUGGEST TRAINING OR ENGAGEMENT GAPS
-

DASHBOARD PURPOSE

- PROVIDES AN OVERVIEW OF ABSENTEEISM TRENDS AND PATTERNS
- HELPS IDENTIFY ROOT CAUSES AND HIGH-IMPACT EMPLOYEE SEGMENTS
- SUPPORTS DATA-DRIVEN HR AND MANAGEMENT DECISION-MAKING

POWER BI DASHBOARD



KEY METRICS

- TOTAL ABSENTEEISM HOURS: 5,124 HOURS
- RISKY LIFESTYLE CASES (E.G., ALCOHOL/SMOKING): 20 EMPLOYEES

CHART 1: TRANSPORTATION EXPENSE & AGE VS. ABSENTEEISM

- LARGER TRANSPORTATION COSTS AND OLDER AGE GROUPS CORRELATE WITH HIGHER ABSENTEEISM
- SUGGESTS LONG COMMUTES AND AGE-RELATED FACTORS MAY INFLUENCE ABSENCES

CHART 2: BMI VS. ABSENTEEISM HOURS (DONUT CHART)

- BMI VALUES 25, 31, AND 30 SHOW THE HIGHEST SHARE OF ABSENTEEISM
- INDICATES OVERWEIGHT/OBESE INDIVIDUALS CONTRIBUTE SIGNIFICANTLY TO ABSENTEE HOURS
-

CHART 3: HIGH ABSENTEEISM BY BMI

- EXTREMELY HIGH ABSENTEEISM IN BOTH UNDERWEIGHT AND OVERWEIGHT BMI RANGES
- SUGGESTS POOR HEALTH (ON EITHER SIDE) IS LINKED TO INCREASED ABSENCES
-
-

CHART 4: SOCIAL DRINKER VS. ABSENTEEISM

- SOCIAL DRINKERS SHOW OVER 1,000+ MORE ABSENTEE HOURS THAN NON-DRINKERS
- RISK BEHAVIORS COULD BE TIED TO LOWER WORKPLACE ATTENDANCE

POWER BI DASHBOARD

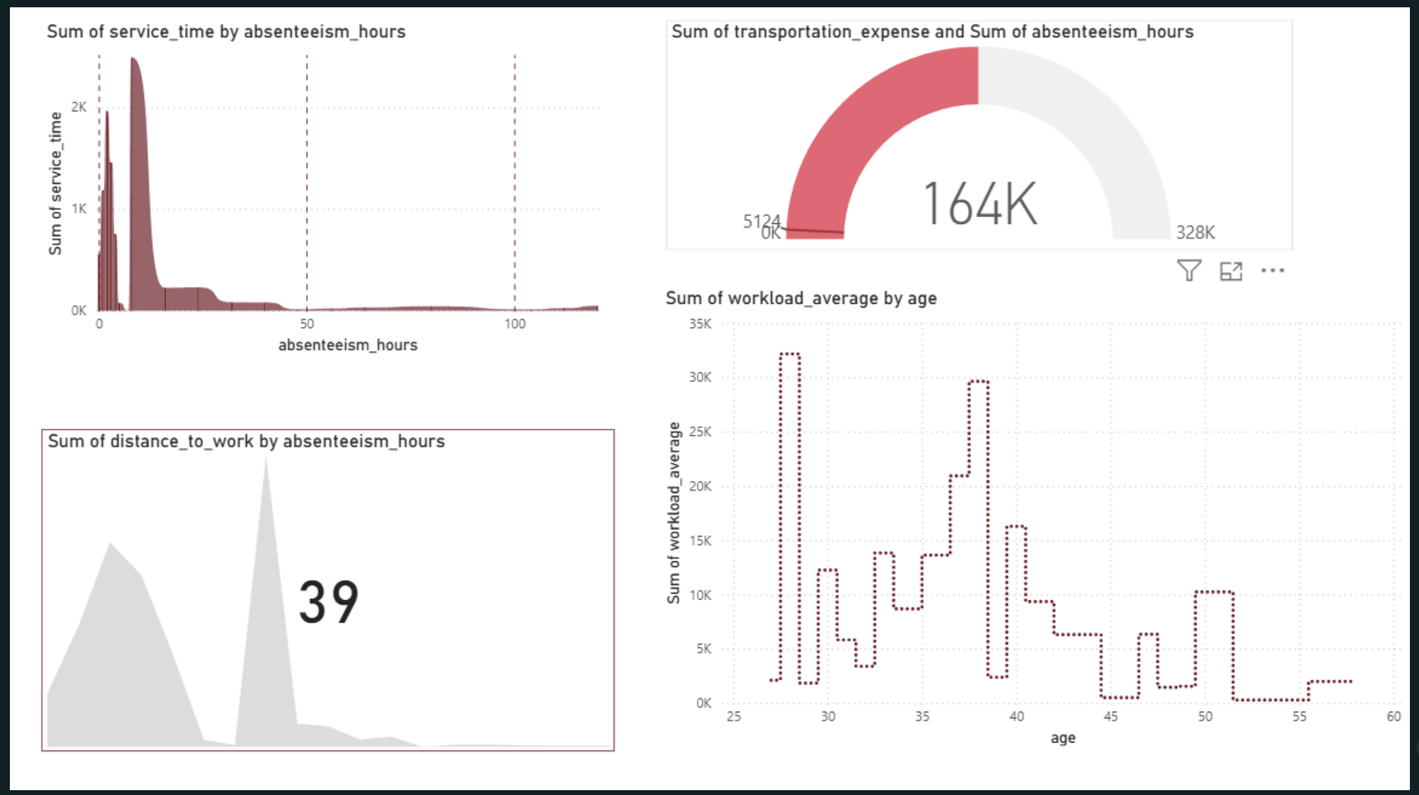


CHART 1: ABSENTEEISM HOURS VS. SERVICE TIME

- MOST ABSENTEEISM IS REPORTED IN EARLY SERVICE YEARS
- FEWER ABSENCES SEEN IN LONG-TENURED EMPLOYEES
-

CHART 2: TRANSPORTATION EXPENSE VS. ABSENTEEISM HOURS

- TOTAL COST IMPACT: 164K
- SUGGESTS ABSENTEEISM MAY LEAD TO SIGNIFICANT LOGISTICAL OR FINANCIAL STRAIN
-

CHART 3: WORKLOAD AVERAGE BY AGE

- HIGHEST WORKLOAD CLUSTERS AROUND AGES 28–38
- THESE AGE GROUPS ALSO SHOW NOTICEABLE ABSENTEEISM, INDICATING BURNOUT RISK

CHART 4: SOCIAL DRINKER VS. ABSENTEEISM

- SOCIAL DRINKERS SHOW OVER 1,000+ MORE ABSENTEE HOURS THAN NON-DRINKERS
- RISK BEHAVIORS COULD BE TIED TO LOWER WORKPLACE ATTENDANCE

POWER BI DASHBOARD

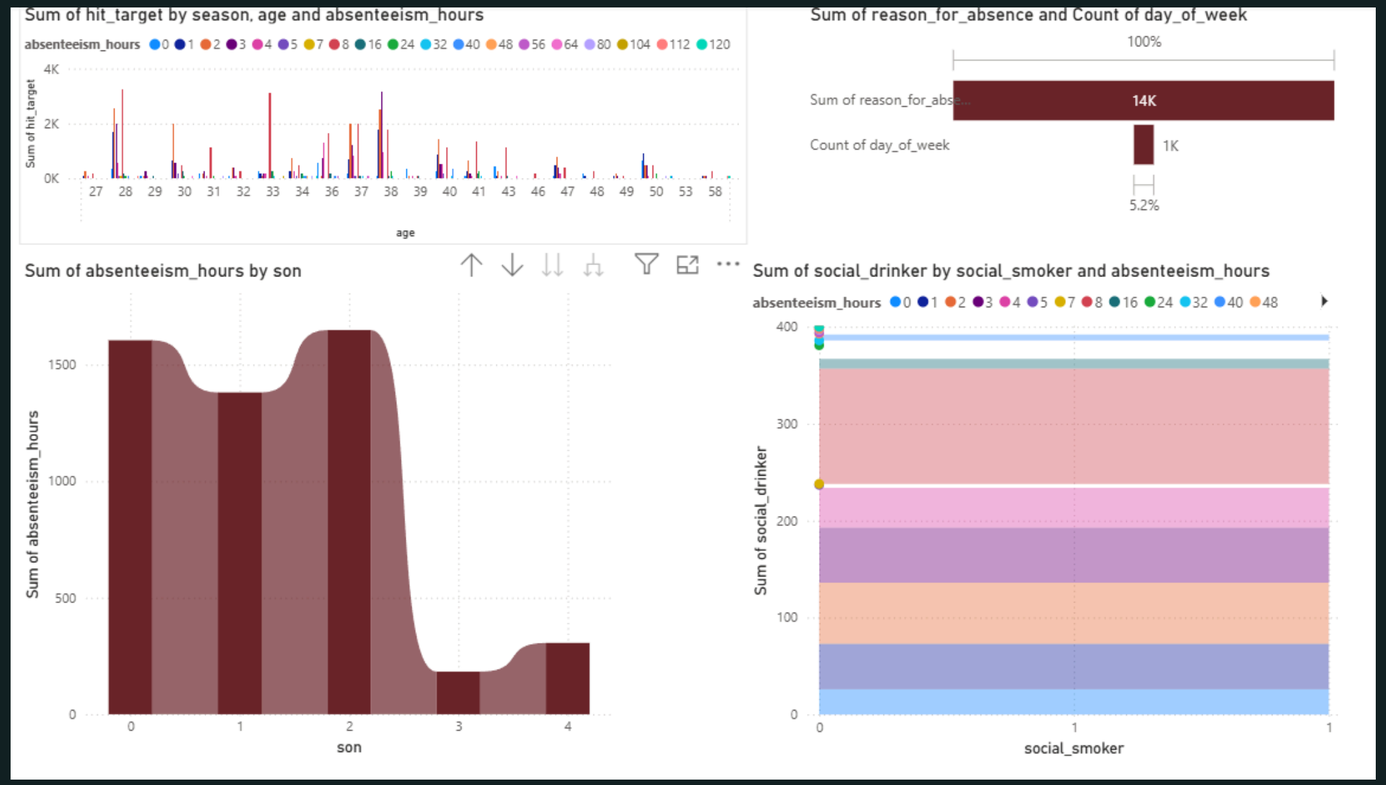


CHART 1: REASON FOR ABSENCE VS. DAY OF WEEK

- ONE PARTICULAR REASON DOMINATES TOTAL ABSENTEEISM WITH 14K TOTAL EVENTS.
- MOST ABSENCES OCCUR MIDWEEK, INDICATING POSSIBLE BURNOUT OR SCHEDULING TRENDS.
- USEFUL FOR PLANNING TARGETED INTERVENTIONS BASED ON WEEKDAY PATTERNS.
-

CHART 2: HIT TARGET BY AGE, SEASON, AND ABSENTEEISM HOURS

- AGES 29–37 SHOW SPIKES IN ABSENTEEISM, WITH VARYING SEASONAL EFFECTS.
- HIGH ABSENTEEISM HOURS (>32) ARE MORE COMMON AMONG YOUNGER WORKERS.
- SEASONALITY AND AGE APPEAR TO INFLUENCE ABSENTEE BEHAVIOR.
-
-

CHART 3: SOCIAL DRINKER & SMOKER VS. ABSENTEEISM

- MAJORITY OF SOCIAL DRINKERS ARE NON-SMOKERS, CONTRIBUTING TO MOST ABSENTEE HOURS.
- MOST ABSENTEEISM OCCURS AMONG SOCIAL DRINKERS, REGARDLESS OF SMOKING STATUS.
- INDICATES SOCIAL BEHAVIORS (LIKE DRINKING) MAY BE LINKED TO ATTENDANCE ISSUES.
-

CHART 4: NUMBER OF SONS VS. ABSENTEEISM HOURS

- EMPLOYEES WITH 0–2 SONS SHOW THE HIGHEST ABSENTEEISM (~1600 HOURS).
- ABSENTEEISM SHARPLY DECLINES WITH 3 OR MORE SONS.
- POSSIBLE LINK TO FAMILY RESPONSIBILITY MANAGEMENT OR TIME CONSTRAINTS.

RECOMMENDATIONS BASED ON ANALYSIS

1. Promote Employee Wellness Programs

- What? High absenteeism is linked to poor health indicators (e.g., high BMI, lifestyle choices).
- Why? Poor health leads to more frequent medical absences and reduced productivity.
- What to do? Offer gym memberships, annual health check-ups, mental health days, and wellness workshops.
-

2. Prepare for Seasonal Absence Peaks

- What? Absenteeism increases in winter months (especially Jan–March).
- Why? Seasonal illnesses and weather conditions affect employee attendance.
- What to do? Implement flexible work policies, allow remote work, and schedule buffer staffing in critical months.

3. Support Employees with High Commute Costs

- What? Employees with longer or costlier commutes show higher absenteeism.
- Why? Commuting stress and transportation barriers lead to missed workdays.
- What to do? Provide transport allowances, promote carpooling, or offer hybrid work schedules.

4. Target Common Absence Reasons

- What? Most absences are due to musculoskeletal, gastrointestinal, and respiratory conditions.
- Why? These may be preventable or manageable with workplace support.
- What to do? Improve office ergonomics, provide healthcare partnerships, and educate employees on preventive care.

THANK YOU